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Exercise : Python Decorators

Step 1

Create an Eclipse Project named `exercise5` and a Python module `calculator5.py`. Copy the contents of the `calculator4.py` from the previous exercise to `calculator5.py`. Also copy the calculations package. You will use the solution code of the previous exercise as a starting point.

Step 2 : benchmark decorator

Now create the following benchmark decorator function in the `calculations.py` :

```
def benchmark(func):
    """
    A decorator that prints the time a function takes
    to execute.
    """
    import time
    def wrapper(*args, **kwargs):
        t = time.clock()
        res = func(*args, **kwargs)
        print (func.__name__, time.clock()-t)
        return res
    return wrapper
```

Add this decorator to the functions in `calculations.py` with the following annotation :

```
@benchmark
```

Run the code and observe the result.

Step 3 : logging decorator

Now create the following logging decorator function in the `calculations.py` :

```
def logging(func):
    """
    A decorator that logs the activity of the script.
    (it actually just prints it, but it could be logging!)
    """
    def wrapper(*args, **kwargs):
        res = func(*args, **kwargs)
        print (func.__name__, args, kwargs)
        return res
    return wrapper
```

Also add this decorator to the functions in `calculations.py` with the following annotation :

```
@logging
```

Run the code and observe the result.

Step 4 : counter decorator

Finally create the following counter decorator function in the `calculations.py` :

```
def counter(func):
    """
    A decorator that counts and prints the number of times a function has
been executed
    """
    def wrapper(*args, **kwargs):
        wrapper.count = wrapper.count + 1
        res = func(*args, **kwargs)
        print ("{0} has been used: {1}x".format(func.__name__,
wrapper.count))
        return res
    wrapper.count = 0
    return wrapper
```

Also add this decorator to the functions in `calculations.py` with the following annotation :

```
@counter
```

Run the code and observe the result.